

Finding Angle Measures of Polygons

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The **interior angles** of a polygon are the angles inside the polygon.

Sum of Interior Angles

The sum of the interior angles of an n -sided polygon is:

$$(n - 2)(180^\circ)$$

Regular Polygon

Each interior angle of a **regular polygon** is:

$$\frac{(n - 2)(180^\circ)}{n}$$

Exterior Angles

The sum of the exterior angles of any polygon is always:

$$360^\circ$$

Each exterior angle of a regular polygon is:

$$\frac{360^\circ}{n}$$

Examples

Example 1

Find the sum of the interior angles of a pentagon.

$$(5 - 2)(180^\circ) = 3(180^\circ) = 540^\circ$$

Example 2

A regular hexagon has equal interior angles.
Find each interior angle.

$$\frac{(6 - 2)(180^\circ)}{6} = 120^\circ$$

Example 3

Find the measure of each exterior angle of a regular nonagon.

$$\frac{360^{\circ}}{9} = 40^{\circ}$$

Practice Exercises

Practice 1

Find the sum of the interior angles of a decagon.

Practice 2

Find the measure of each interior angle of a regular octagon.

Practice 3

Find the measure of each exterior angle of a regular dodecagon.

Practice 4

One exterior angle of a regular polygon measures 45° . How many sides does the polygon have?

Practice 5

One exterior angle of a regular polygon measures 30° . How many sides does the polygon have?

Practice 6

The interior angles of a quadrilateral are 95° , 88° , and 102° . Find the fourth angle.

Practice 7

The interior angles of a pentagon are 110° , 125° , 118° , 132° , and x° . Find x .

Activities

Activity 1

Find the sum of the interior angles of a heptagon.

Activity 2

Find the sum of the interior angles of a nonagon.

Activity 3

A regular pentagon has equal interior angles.
Find each interior angle.

Activity 4

Find the measure of each interior angle of a regular decagon.

Activity 5

Find the measure of each exterior angle of a regular octagon.

Activity 6

Find the measure of each exterior angle of a regular pentadecagon.

Activity 7

One exterior angle of a regular polygon measures 60° . How many sides does the polygon have?

Activity 8

One exterior angle of a regular polygon measures 24° . How many sides does the polygon have?

Activity 9

The interior angles of a quadrilateral are 85° , 97° , and 108° . Find the fourth angle.

Activity 10

The interior angles of a pentagon are 115° , 128° , 121° , 136° , and y° . Find y .

Thank you!